TEXORA · LECTURE SHEET

Recipe Prediction from a Given Sample

A buyer hands over a dyed swatch and asks for ten thousand metres of it. The colorist's first job is to work backwards — from the colour on the sample to the dyes and quantities that will reproduce it.

WP-3207 · ADVANCED MACHINERIES AND RELEVANT CALCULATIONS · TOPIC 1 · SPECIALIZATION — WET PROCESSING

1. What "recipe prediction" means

Recipe prediction is the estimation of the dye combination and concentrations needed to reproduce a given standard (the sample) on a specified substrate, before any production dyeing takes place. It answers the buyer's sample with a *starting recipe* — close enough that one or two laboratory corrections bring it inside the agreed tolerance, typically $\Delta E \leq 1.0$ under the specified illuminant.

Prediction is needed because colour is not a property of a dye alone: the same dye gives different depths on different fibres, at different liquor ratios, and in the presence of different auxiliaries. The prediction method must therefore be built on dyeings made **on the same substrate, by the same process** as the production it feeds. A recipe predicted from polyester calibrations is worthless for cotton.

2. The information carried by the sample

- **Substrate:** fibre type and blend ratio, construction (woven/knit), and the preparation state — a sample dyed on well-bleached fabric will not match on under-scoured goods at the same recipe.
- **Colour:** measured instrumentally as spectral reflectance $R(\lambda)$, usually 400–700 nm at 10 or 20 nm intervals, or received as a physical swatch to be measured in the laboratory.
- **Dye class:** inferred from the fibre and the fastness demands — reactive dyes for cellulosics with wash-fastness requirements, disperse for polyester, and so on. Simple spot tests and the sample's history narrow the class before matching begins.
- **Target conditions:** the illuminant(s) for approval (D65 first, then TL84 and A as metamerism checks), the tolerance, and the production machine and liquor ratio the recipe must serve.

3. The calibration (build-up) series — the heart of the method

For each candidate dye, the laboratory dyes a **build-up series** on the production substrate: typically 0.1, 0.25, 0.5, 1.0, 1.5, 2.0, 3.0 and 4.0% owf, all at fixed salt, alkali, temperature and time. Each dyeing is measured, and its colour strength is expressed as **K/S** from the Kubelka–Munk relation:

$$K/S = (1 - R)^2 / 2R$$

where R is the fractional reflectance at the wavelength of maximum absorption. K/S rises nearly linearly with concentration at low depths and bends over as the fibre saturates — the curve is the dye's fingerprint on that substrate. For example, R = 0.18 at λ_{max} gives K/S = $(0.82)^2/(2 \times 0.18)$ = 0.6724/0.36 = **1.87**.

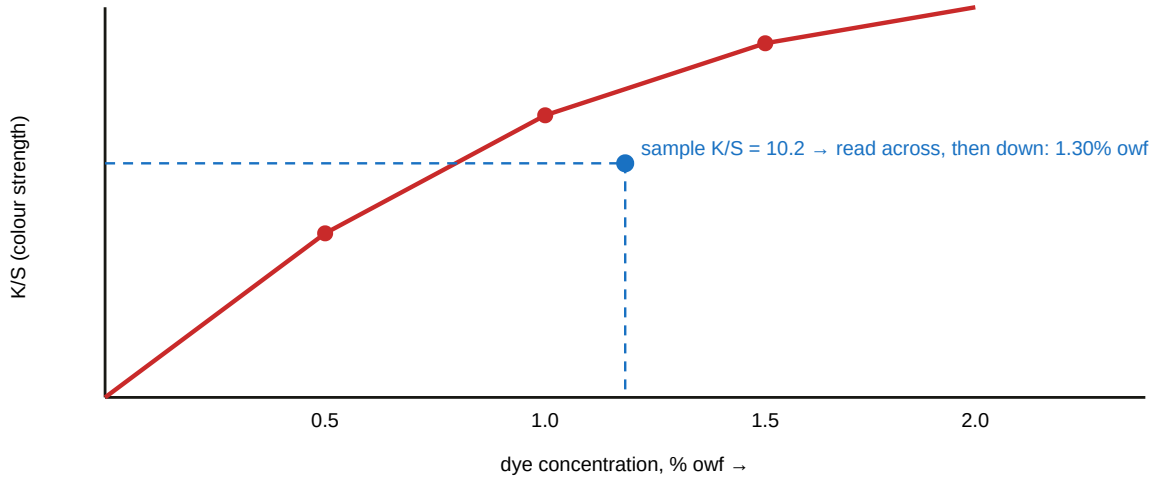


Fig. 1 — Build-up curve of a single dye on the production substrate. The sample's measured K/S is projected onto the curve to read off the predicted concentration (worked example below).

4. Fully worked prediction — single dye

WORKED EXAMPLE 1 — DEPTH MATCHING FROM THE BUILD-UP CURVE

A reactive dye on scoured–bleached cotton gives this calibration (K/S at λ_{max}): 0.5% → 4.2; 1.0% → 8.1; 1.5% → 11.6; 2.0% → 14.8. The buyer's sample measures K/S = 10.2.

The sample lies between the 1.0% and 1.5% dyeings. Interpolate: fraction = (10.2 − 8.1)/(11.6 − 8.1) = 2.1/3.5 = 0.60. Predicted concentration = 1.0 + 0.60 × 0.5 = **1.30% owf**.

Bulk requirement for a 250 kg lot: dye = 1.30% × 250 kg = **3.25 kg**. If the plant's experience factor for lab-to-bulk depth loss is 5%, the corrected recipe is 1.30 × 1.05 = 1.365% owf → 3.41 kg. The correction factor belongs to the individual plant and machine; it is earned from records, not assumed.

Interpolation is legitimate only between adjacent calibration points and only in the near-linear region. Above the bend of the curve, depth becomes insensitive to concentration and prediction errors explode

— a practical reason deep shades are matched in smaller steps.

5. From one dye to three — the prediction sequence

1. **Measure the standard:** reflectance curve of the sample on a spectrophotometer, averaged over several positions and both fabric directions.
2. **Select the dye set:** the smallest set (usually a trichromacy — yellow, red, blue) from the correct dye class whose combined gamut contains the sample colour and whose exhaustion/fixation profiles are compatible.
3. **Predict by additivity:** Kubelka–Munk K/S values are approximately additive over dyes, so the mixture's K/S at each wavelength is the sum of the components' contributions. The computer match-prediction system solves for the concentrations that reproduce the standard's K/S spectrum — the full method is Topic 3.
4. **Dye and measure the trial:** a laboratory dyeing at the predicted recipe on the true substrate; compute ΔE against the standard.
5. **Correct:** adjust the component responsible for the residual error — hue errors by swapping weight between two components, depth errors by scaling all three — and re-dye. One or two iterations should land inside tolerance.
6. **Freeze the recipe:** record dyes, % owf, salt/alkali, LR, temperature profile and the measured ΔE under each illuminant. The frozen recipe — not the swatch — is what production receives.

6. Sources of prediction error

ERROR SOURCE	EFFECT AND CONTROL
Substrate difference	Different preparation, blend or lot of base fabric changes uptake. Always calibrate and trial on production fabric from the same preparation route.
Dye lot variation	Commercial dye strength varies between lots; standardise each lot against the retained standard and correct the recipe by the strength ratio.
Liquor ratio shift	Exhaustion equilibria depend on LR; a recipe predicted at 1:20 does not transfer unchanged to a 1:8 jet (see Topic 2).
Water quality	Hardness and metal ions dull some shades; use the plant's process water in lab dyeings, with the production sequestering agent.
Metamerism	A match under D65 may fail under TL84 if the dye set differs chemically from the standard's. Predict with the same dye class as the standard whenever possible and check all specified illuminants.

7. Glossary — key terms

- **Build-up series:** dyeings of one dye at increasing concentrations used to calibrate colour strength against concentration.
- **K/S:** Kubelka–Munk ratio of absorption to scattering coefficients; the working measure of depth.
- **Standard:** the approved sample whose colour is to be reproduced.
- **Metamerism:** two colours matching under one illuminant but not another because their spectra differ.
- **Experience factor:** a plant-specific correction between laboratory prediction and bulk result.

8. Key points

- Prediction is calibration on the real substrate plus disciplined interpolation — never guesswork from dye names.
- K/S converts a measured colour into arithmetic; the build-up curve converts arithmetic back into a recipe.
- Every prediction is provisional until a lab trial on production fabric confirms ΔE inside tolerance under all specified illuminants.

9. Summary

Recipe prediction turns a buyer's swatch into numbers (reflectance, K/S), numbers into a candidate recipe (calibration curves and additivity), and the candidate into a frozen production recipe through measured correction. The method's accuracy is set long before the trial dyeing — by the quality of the calibration series and the honesty of the plant's correction factors.

10. Practice

1. A dye's build-up on cotton gives K/S: 0.5% → 3.9; 1.0% → 7.4; 1.5% → 10.5; 2.0% → 13.1. A sample measures K/S = 9.0. Predict the concentration and the dye needed for a 180 kg lot.
2. Calculate K/S for a dyeing whose reflectance at λ_{max} is 0.09.
3. Why must the build-up series be dyed on the same substrate and preparation as production, rather than taken from the dye maker's pattern card?
4. A predicted recipe matches under D65 (ΔE 0.6) but fails under TL84 (ΔE 2.3). Name the phenomenon and two remedies.

Answer key

1. Sample K/S 9.0 lies between 1.0% (7.4) and 1.5% (10.5): fraction = (9.0 − 7.4)/(10.5 − 7.4) = 1.6/3.1 = 0.516 → 1.0 + 0.516 × 0.5 = 1.26% owf. Dye for 180 kg = 1.26% × 180 = 2.27 kg.
2. K/S = $(1 - 0.09)^2/(2 \times 0.09)$ = 0.8281/0.18 = 4.60.
3. Colour strength per unit dye depends on the fibre, its preparation and the dyeing process: the pattern card uses the maker's standard substrate and conditions, so its K/S-per-% relation does not hold on the plant's fabric — predictions made from it carry a systematic error that no later correction fully removes.
4. Metamerism. Remedies: re-match using the same dye class (ideally the same dyes) as the standard, and select among candidate recipes the one with the lowest metamerism index across all specified illuminants, accepting a slightly larger D65 ΔE if necessary.

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